

Network Support

Network support is not necessary for home users, but since we're also considering small offices, this could be a decisive feature. The Brother HL-5170DN is the only printer in our comparison that has Ethernet installed; it is an optional feature on the Canon LBP2900, meaning you can buy a compatible network card and install it if and when you feel the need.

The Driver Interface

Drivers are necessary to install and run the printer. But a driver is also an interactive tool for setting up print jobs and more.

We found the Epson EPL 6200L's driver a little awkward. It didn't malfunction, but it doesn't have settings for Normal, Draft and Best. In sharp contrast, the Canon LBP2900 has an over-friendly driver interface. It offers printing modes such as 'Standard', 'Text/Lines', and even 'Web Pages'. The Canon LBP2900's driver interface includes a progress indicator—in case of an error, the interface either stalls the print job or cancels it, and issues an error message on the screen.

Performance

Figures for the sub-sections under performance—speed and quality—



Brother HL-5170DN

have been mentioned in the comparison table for monochrome lasers so you can get a better idea of how each printer performed.

You might notice that neither of our winners lead in the performance department; also, there isn't much difference between the two. The Samsung ML-2010 ranks second—next only to the Canon LBP2900—in the performance department.

Quality

The stated resolution is an important aspect when judging a printer, but the best way to know about a printer's print quality is by actually comparing printouts by various printers. This is not

always possible, as outlets normally do not let you take demo printouts. Our performance analysis therefore concentrates on real-world tests, where we use different types of test documents containing graphs, textures, varying point-size text, images and colour gradients.

One of the key discriminators in the tests was white text on a black background with varying text sizes. The Canon LBP2900 and the Brother HL-5170DN produced good contrast, and hence scored better. The Canon LBP2900 topped in the quality analyses by producing much better prints of our combi-document. Apart from the Xerox



Samsung ML-2010

How We Tested

The Test Bed

The test bed we used comprised an AMD 64 3000+ (socket 754) processor fitted on an MSI K8T Neo motherboard. The combination was well supported by 1 GB of DDR 400 RAM and an nVidia GeForce 5950 GT 128 MB graphics card. The hard disk was a Maxtor 120 GB SATA, and the OS was Windows XP with SP1. The system was loaded with the latest chipset and graphics drivers. We connected the sample printers to our test bed via the USB interface. To derive optimum performance, the USB 2.0 patch was loaded onto our Windows installation. Berga (A4 size) paper served as the medium for printing.

Features

The many parameters we considered were weighted differently. Parameters such as resolution, memory, network support, etc. were given more importance.

Performance

Performance was broadly categorised under speed and quality. The speed tests were done by timing the printer from the instant we clicked on the Print button in the Print dialog box to the time the page dropped out onto the output tray. These tests were carried out using different types of source files such as a regular text document, a combi-document (containing text, images, concentric circle patterns, a colour gradient bar, etc.) and a business type document that had graphs, logos and photos apart from text. A full-page photo printout was timed for the colour printers.

Under Quality, we looked at certain distinguishing patterns in each of our source files; for instance, in the photo printout test, where we used a photograph of a face, not every printer could properly reproduce skin colour and wrinkles under the eyes and forehead. The printers were rated on how well they could reproduce such details. The quality of the printouts from the mono laser printers was judged mainly on the basis of text, graphs, concentric pattern reproduction, and the quality of white text on a black background.

Price

The price of the toner was given more consideration than the cost of the printer, but that's not to say that the cost of the printer didn't affect the results.

Narrowing Down The Winners

The overall score is based on the scores obtained from the Features, Performance and Price segments. These segments are again weighted over each other, keeping in mind the typical home or small office user. Performance was rated over features and price—price was given the lowest weightage in the tests for both colour and mono printers. That said, price was given relatively higher consideration in the case of mono laser printers, because those who opt for colour laser printers would be more concerned about performance and features.

We chose the winners based on the final score obtained upon applying all three parameters.